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Toward an Understanding of Dynamic Moral Decision Making: Model-Free and Model-Based Learning

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Abstract In business settings, decision makers facing moral issues often experience the challenges of continuous changes. This dynamic process has been less examined in previous literature on moral decision making. We borrow theories on learning strategies and computational models from decision neuroscience to explain the updating and learning mechanisms underlying moral decision processes. Specifically, we present two main learning strategies: *model-free* learning, wherein the values of choices are updated in a trial-and-error fashion sustaining the formation of *habits* and *model-based* learning, wherein the brain updates more general cognitive maps and associations, thus sustaining flexible and state-dependent behaviors. We then summarize studies explaining the neuro-computational processes of both learning strategies—the calculation of prediction errors and valuation. We conclude by emphasizing how the incorporation of dynamic aspects in moral decision making could open new avenues for understanding moral behaviors in a changing world.

Keywords Corruption · Bribery · Decision making · Decision neuroscience · Learning · Model-based

Abbreviations

DN	Decision neuroscience
RPE	Reward prediction error
SPE	State prediction error
VMPFC	Ventro-medial prefrontac cortex
SMH	Somatic marker hypothesis
DLPFC	Dorso lateral prefrontal cortex

Learning processes begin at birth and last throughout an individual's life, reinforced by environmental demands, daily experiences and observations (Cole and Packer In press). The current paper aims to understand the dynamic updating process of individuals' moral decision making, given that experiences and observations vis-à-vis environmental demands can lead to revisions of moral priorities. This learning mechanism is manifested in many ways. Ethical training and interventions harness the notion that moral systems are malleable and can be changed. Moral values could drastically change when the environment changes. For instance, homicide might be impermissible to a person during peace; yet that very same person might accept or even value the act during war. Moreover, situational factors stemming from organizational cultures, industrial norms, and national cultures also impose challenges for decision makers' capability in updating their moral decision making (Christopoulos and Hong 2013; Hong 2008). We aim to delineate how Decision

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Neuroscience can shed light on the dynamic processes of moral decision making.

Previous theoretical models and research efforts in morality and Business Ethics have successfully detailed factors influencing moral decision making. These conceptualizations have already hinted at dynamic processes updating choices. For instance, Jones' issue-contingent model (Jones 1991) underlines the significance of factors that typically support almost any kind of decision: magnitude of consequences, social consensus, probability of effect, temporal immediacy, proximity, and concentration of effect (Jones 1991). To the extent that these factors vary across situations, the question of how individuals make moral decisions contingent on different contexts arises. Moreover, different cultures also shape the contents and priorities of moral bases, thus giving rise to differences in the intuition of moral right versus wrong in different societies (Haidt 2001).

Reynolds' neurocognitive model (Reynolds 2006) brings forward the issue of updating in moral decision making. This is reflected in the feedback loops characterizing all three sub-processes (Reynolds 2006; see Fig. 1): prototypes are searched and updated based on the structuring of information; moral rules are updated based on continuous restructuring of processes, based on either rationalization or active judgment. This neurocognitive model also emphasizes that humans do not only assess the environment and update their actions, but they also assess and update their own **beliefs** (that is, their *models*) about the environment. As it will be detailed below, in a similar vein, decision neuroscience (DN) focuses on this metacognitive element—the updating of models of the state of the world in dynamic environments—to explain how organisms adapt to their environment.

Furthermore, another aspect DN could contribute is to uncover the computational and neurobiological basis of how individuals compare expectations with reality. Previous theories have examined how decision making can be shaped by previous states. For example, control theory (Carver and Scheier 1982) suggests that moral identity can act as a self-regulatory mechanism that drives moral behaviors in order to maintain one's identity as a moral person (Mulder and Aquino 2013, p. 220). Individuals make moral decisions dynamically, depending on previous choices. In fact, this mechanism is the cornerstone of a diverse set of learning behaviors, ranging from pure reinforcement learning to learning from others and even updating beliefs of the world.

In the present paper, we provide a DN-based framework to explain how moral decisions change as a function of time, individuals' previous choices and feedback mechanisms (as illustrated in Fig. 1), and to identify the biological basis of these updating mechanisms. Previous theories mostly focused (though not exclusively) on how moral choices are made and how outcomes are evaluated. To supplement this approach, the present paper focuses on how moral learning and changes can be achieved through feedback and updating, grounded in a computational approach. We will review the corresponding behavioral and neurological evidence throughout the present paper.

It is important to note that the decision-making processes and mechanisms described throughout the present paper ***are not necessarily happening at a conscious, planned, and deliberate level***. Rather, DN posits that these steps have arisen as a *computational necessity* for an information processing system to reach a decision. These functions and processes could, and to a great extent, do happen before any decision is made and could very well happen mostly without reaching consciousness. Even when deliberate processes are employed, a significant part of the supporting computations could occur in a non-deliberate fashion. Here, the reader should be careful to dissociate the word 'computation' from deliberation. For instance, associations between stimuli and their predicted values or even bodily reactions could be interpreted as intuitive—yet they also still need to be sustained by some form of

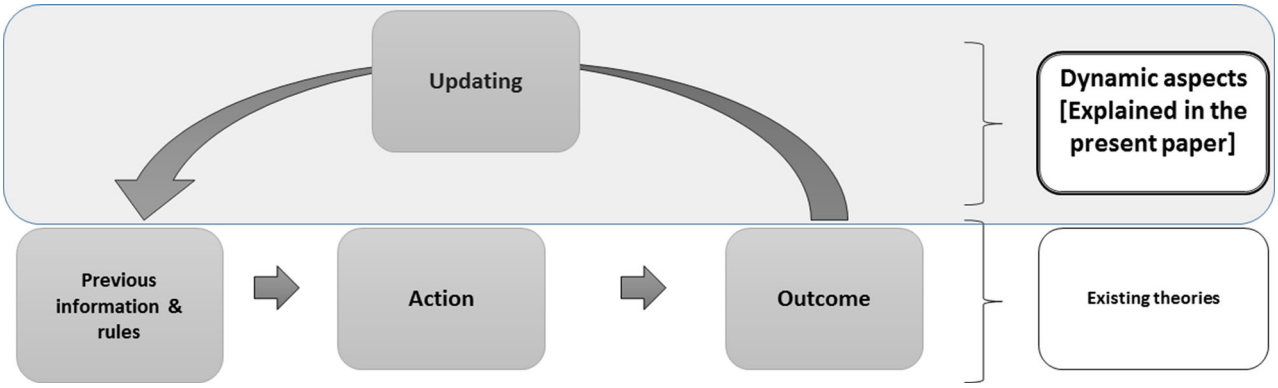

Fig. 1 An abstract representation of a decision-making process

computation, and this is what DN aims to clarify. Thus, it should be clear that, for DN, the decision-making process need not be totally conscious, calculative, and purposeful.

Second, our approach here is descriptive—and not normative. In other words, the question here is not what people *ought* to do (which is more a question of traditional philosophy dating back to Kant, Hume, Mill and even Aristotle), but what people *actually* do. The natural extension of this is to identify how external conditions impact behaviors, how individual differences (including biological constraints) influence such decisions, and the mechanisms preceding moral decisions (cf. Young and Koenigs 2007).

In the following sections, we first briefly introduce Decision Neuroscience. Subsequently, we introduce the two main learning strategies that the brain employs to predict the environment (model-free vs. model-based learning), the valuation processes, and how these mechanisms can be used to understand moral decision making. We conclude by discussing how Decision Neuroscience could be integrated to further our understanding of moral decision making and then identify some areas for future research.

An Overview of Decision Neuroscience

We first provide an overview of DN regarding the major conceptual foundations and contributions, with a focus on decisions in dynamic environments. We summarize the core concepts and definitions in Table 1. The goal of Decision Neuroscience is to understand the behavioral, computational, and neurological underpinnings of decision parameters and how these are integrated and updated in the human brain. For instance, understanding the mechanisms underlying valuation (the process of attaching values to different options via a learning process) or risk are central questions in Decision Neuroscience (Christopoulos et al. 2009; Delgado et al. 2005; Knutson et al. 2008; Kuhnen and Knutson 2005; Plassmann et al. 2007; Platt and Huettel 2008; Preuschoff et al. 2008). DN offers a systematic framework that comprehensively represents the mechanisms involved in the decision-making process. The fundamental assumption of DN's approach is that decision making can be computationally and algorithmically described. The framework has received not only strong theoretical support, as inspired by economics, game theory, and psychology, but also significant confirmatory biological evidence (Bossaerts 2009; Camerer 2008; Glimcher and Fehr 2013; Rangel et al. 2008; Schultz 2008).

The foundations of Decision Neuroscience could date back to Ivan Pavlov (Pavlov 2003) who first examined the physiological processes of orienting attention to rewarding stimuli and learning to assign values to originally neutral stimuli. Moving forward, one of the major breakthroughs of neuroscience and cognitive science was the understanding that processes—such as attention, memory, and language—are not unitary processes but can actually be broken down to more basic processes. This is especially the case for decision making, which involves many stages, including identification of options, valuation, choice, action, and updating. It is essential to parameterize these stages; for instance, valuation can be specified into several parameters, such as value, risk, and time, as well as social and moral considerations. This parameterization of decision-making processes emerges as a necessary way to model and investigate the neuroscientific evidence (Montague and King-Casas 2007). Algorithmic representations (Daw and Doya 2006) and the related computational fitting of neural and behavioral observations (Dayan and Daw 2008; Montague et al. 2012) further clarify the decision-making process.

DN has thus reached the point at which a concrete program has been established. Examples of important questions include: the exact mechanisms through which people learn and update their actions in uncertain and dynamic environments; the identification of the basic decision parameters and how they are represented in the human brain and body; the role of physiological/emotional/bodily responses (heart rate, skin conductance and breathing); how social aspects such as trust, reciprocity, cooperation, social influence or even architectural parameters are integrated (Delgado et al. 2005; Fehr et al. 2005; King-Casas et al. 2005; Krueger et al. 2007; Roberts et al. 2016); how different decision parameters are integrated to allow decision and action. This research has highlighted the importance of understanding the psychological and biological foundations that could also characterize theories and models of organizational behavior (Decety et al. 2004; Rilling et al. 2002).

Understanding Dynamic Moral Decision Making: Model-Free and Model-Based Learning

As noted above, decision makers facing ethical issues experience a dynamically changing environment. Mobility in job and residence impose challenges of adapting to different criteria associated with moral decision making. In fact, existing theoretical approaches (Haidt 2001, 2007; Jones 1991; Reynolds 2006) have already emphasized the importance of understanding the *dynamic* aspects of moral decision making. Uncovering the exact mechanisms would need to incorporate (i) value-based learning, where outcomes and probabilities are used to guide decisions, (ii) feedback processes that take into account previous choices

Table 1 Decision neuroscience: basic concepts and short explanations

Temporal discounting	The behavioral tendency to prefer rewards with smaller value delivered earlier (compared with larger value rewards delivered later); it is related to impulsivity
Risk attitudes	Preferences over risky alternatives, ranging from risk-averse to risk-seeking. Here, risk is treated as a generic terms that includes ambiguity (i.e., events with unknown probabilities) as well as different facets of risk, such as variance, skewness, and probability
Dopamine	A neurotransmitter (i.e., a chemical that subserves communication between neurons). Very strong evidence suggests that dopamine neurons support reinforcement learning and valuation, especially via prediction errors. In humans, striatum (nucleus accumbens) is an area that is strongly associated with dopaminergic activity
Prediction error	A basic computation that compares what was expected with what was actually delivered. In a formal computational model (TD-Learning), this computation allows the updating of values attached to different stimuli, which in turn enables learning. Within model-free learning, the computations suggest the generation of Reward Prediction Errors; within model-based learning, the computations involve the generation of State Prediction Errors (comparing expected and realized states)
Utility	A concept inspired by microeconomics and Game Theory suggesting that decisions can be described "as if" the decision maker is assigning a subjective value to each alternative, thus establishing a common currency by way of which all alternatives can be compared. It should be differentiated from the way the term Utility is used by 'Utilitarian' moral philosophy (see below and in text)
Somatic marker hypothesis	The hypothesis that bodily responses, such as heart rate or skin conductance, can act as "markers" of past experience, thus informing present responses
Utilitarian choice (morality)	A moral school of thought suggesting that the basic criterion of morality is to choose actions that increase the overall utility ("the greater good"), even if they violate basic principles
Deontological choice (morality)	A moral school of thought suggesting that the basic criterion of morality is to choose according to basic, widely accepted principles (such as not committing murder), even if they decrease overall utility

and outcomes, thus allowing adaptive changes, and (iii) different levels of learning—i.e., processes that update values, actions as well as beliefs about the world.

Remarkably, research in Decision Neuroscience has already identified the central basic mechanisms underlying dynamic decision making. The key processes identified are ***valuation*** and the generation of ***prediction errors***. Figure 2a explains the computational framework for such a decision process in an abstract form. Figure 2b illustrates a very simplified but concrete decision process in business ethics, where valuation and feedback loops (prediction errors) exist. Specifically, *valuation* is the process through which a value is assigned to an action (Rangel et al. 2008), e.g., to bribe or not to bribe, whereas a *prediction error* is the difference between the realized and expected outcome. For example, when making a decision regarding whether or not to give a bribe, decision makers would need to (implicitly) assign a value to each action option and act

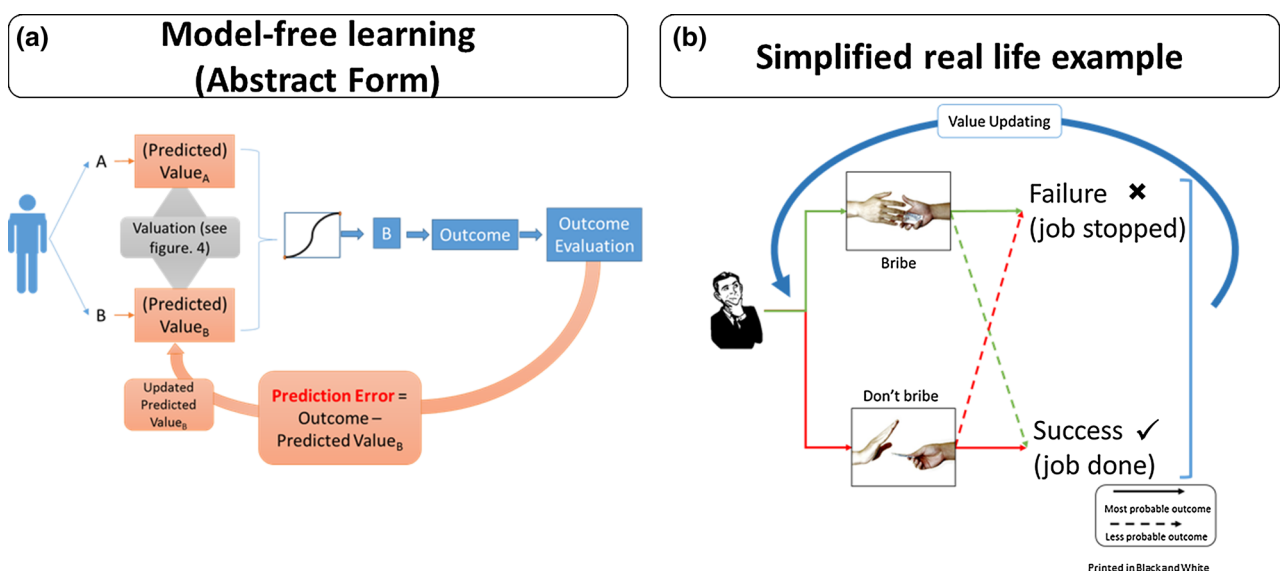

Fig. 2 Model-free learning: abstract form and a real-life example

accordingly (usually the option with the highest subjective value would be chosen). The valuation is subsequently used in the estimation of the prediction error, which compares the predicted value of the chosen option (assigned during the decision stage) with the actual value of the outcome (realized during the outcome stage). Assume, for instance, that a decision maker might expect that bribing would yield a positive outcome; yet, after s/he bribes, s/he is instead punished. This would constitute a ***negative prediction error***, as the difference between realized and expected outcome is negative. The prediction error, in turn, is used in the valuation stage of the next decision. The decision maker has now a new estimation of the predicted value of the option because of the previous outcome. Therefore, within a decision cycle, valuation is used to compute the prediction error; in the next cycle, the prediction error updates the valuation of a particular option. Going through multiple trials of decision making and receiving feedback of the outcomes associated with the choices, valuation would occur with higher precision and the prediction errors would become smaller in magnitude, assuming that the context is relatively stable.

Epistemological Predecessors and Related Dual Approaches

Widely examined in physiology, animal behavior, and neuroscience, there are two main epistemological streams explaining choice and learning in dynamic environments. The first one, reinforcement learning, is based on an undemanding learning rule, the Law of Effect (Thorndike 1911), stating that if a response to a stimulus is rewarded (or punished), then the connection between the stimulus and the response is strengthened (or weakened). The basic premise of reinforced learning includes automatic and semi-automatic responses (e.g., salivation) toward basic rewards (e.g., food) and punishments (e.g., electric shock), along with the establishment of similar responses to stimuli that by association predict these rewards and punishments.

Apparently, this type of learning is not successful in explaining and establishing highly complex behaviors in demanding environments. Indeed, following Thorndike's proposition, Tolman (1948) proposed that responses could be planned based on an internal "cognitive map" representing contingencies. In this case, individuals instead of using—sometime tedious—computations, use a look-up table by storing information every time an action is taken and a new state is observed.

The current epistemological offspring of these two types of learning strategies are the ***model-free (habitual)*** and ***model-based (goal-directed)*** learning strategies (illustrated in Fig. 3 and summarized in Table 2). In this context, the term "model" refers to mental representations of the world. (e.g., internal cognitive maps or decision trees). Model-free learning is a trial-and-error updating mechanism, which is, at least partially, the basis of habit formation, whereas model-based learning is employed for generating goal-directed plans, especially in complex and hierarchical (i.e., environments with multiple dependencies) environments. Research has shown computational, behavioral, and biological evidence indicating that the two learning strategies are distinct but can also be used in a parallel fashion and can be integrated (Daw et al. 2011; Fermin et al. 2010; Glascher et al. 2010; Wunderlich et al. 2012).

Previous efforts in ethical decision making have also adopted the dual processes approach (Illes and Bird 2006; Reynolds 2006). For example, Reynolds' neurocognitive model (Reynolds 2006) was developed based on the dual-processing theory (Lieberman et al. 2002), such that the reflexive system (X-system) represents a predominantly non-conscious process, whereas the reflective reaction (C-system) represents a predominantly deliberative process to respond to a situation. Both systems operate on prototypical representations of ethical principles and situations. A similar distinction (Kahneman and Frederick 2002, 2005) is between System 1 (fast and intuitive) and System 2 (reflective and slow). Model-free and model-based learning strategies are extensions of such dual approaches to the learning domain, with neurological evidence suggesting that humans use both systems but to a different degree (Beierholm et al. 2011).

In the following section, we will introduce each learning strategy and the mechanisms of how the two strategies are integrated. Again, all the processes described are not necessarily deliberative and planned.

Model-Free Learning and Reward Prediction Error (RPE)

Model-free learning is based on simple reinforcement learning rules; yet, these rules are robust enough to explain a very wide range of behaviors. A very simple example could be a decision maker learning that pressing a lever will deliver a can of soda. At least two pieces of information are needed here: first, that pressing the lever is followed by the delivery of a can of soda; second, that the beverage is desirable. However, this might be cognitively inefficient, as the same process can be produced by a simpler mechanism: the decision maker just learns that in the presence of the button he or she needs to press the lever—i.e., stimuli are 'wired' to responses in an almost automatic (or *habitual*) way. This very simple mechanism is the basis of habit formation (for a general review see (Daw 2012)). Building on the principles of associative learning, neurophysiologists, and psychologists expanded

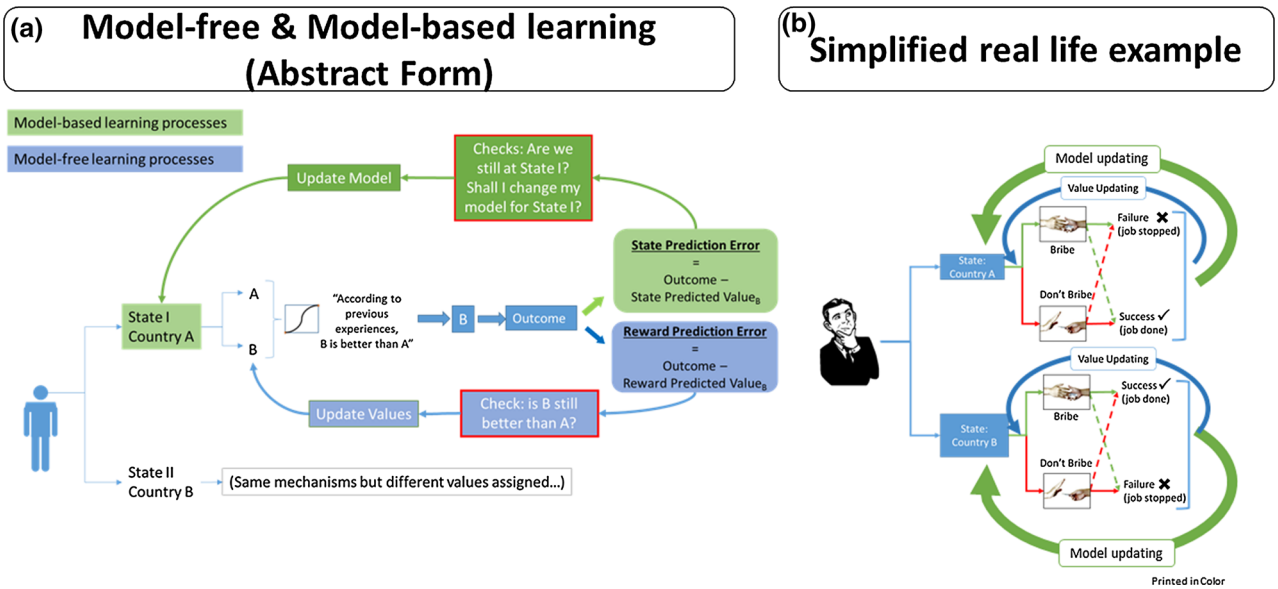

Fig. 3 Model-free and model-based learning

Table 2 Summarized differences between the two learning strategies

Model-free learning	Model-based learning
These responses subserve a wider set of situations that can be acquired by repeated training, usually by trial-and-error. They include semi-automatic and routine behaviors, such as driving the same route every day to work or having a drink when entering a specific bar. More complex behaviors, such as excessive gambling, could be demonstrations of habitual learning. Habitual learning is a slow process, but once habits are solidified, they are resistant to change and can generate quick responses (some of them are *habits*)	Instead of learning associations by trial-and-error, the agent employs a cognitive map (akin to a 'look-up' table or a decision tree) representing associations. Using simpler indexing operations, goal-directed systems assign values to actions (as opposed to habitual learning, which assigns values to outcomes). More complex behaviors, such as how to navigate a new environment or how to respond to a complex social situation, are attended by this system
Main advantages are that, once established, responses (even complex sequences of responses) can be quickly generated, whereas updating is efficient in not very complex environments	Main advantage is flexibility and quicker adaptation
Main disadvantage is that the response is slow and inaccurate when the environment has changed (or has many different states)	Main disadvantage is the need for high memory, and the possibility of following inaccurate models
Similar terms/epistemological antecedents: habitual responses; reflective responses; C-system; System 1	*Similar terms/epistemological antecedents: goal-directed responses; reflexive responses; X-system; System 2*
Hybrid learner	
In reality, both learning strategies are employed, but with different weights. Neuro-computational models suggest that the strategy with less uncertainty will be overweighed; preliminary studies also suggest individual differences	
Recall that processes here need not be deliberative or even conscious, but they could happen rather at a non-conscious level	

by addressing increasingly complex behaviors and environments. Quickly, it was understood that there was a need for a robust computational model that could explain the physiological and behavioral responses. Inspired by computer science and robotics, reinforcement learning (RL) has been repeatedly shown to have a close correspondence with behavioral manifestations of adaptive responses as well as their biological underpinnings. RL, in fact, includes an extensive set of different learning algorithms that aim to improve choices based on the feedback of previous rewards and punishments. In its simpler form, actions (or, in some cases, strategies) that are rewarded will occur with higher frequency, whereas the opposite holds true for actions that are followed by punishment. In sum, the basis of RL is that stimuli and possibly actions that are associated with rewards will acquire significance and be preferred.

Is this associative mechanism somehow represented in the brain? Or is it just a mere theoretical assumption? Evidence suggests that activity of the so-called dopamine neurons can be the basis of such computations. For

instance, dopamine neurons will immediately respond to the delivery of juice (unconditioned stimulus), thus signaling a rewarding outcome (Hollerman et al. 1998; Schultz et al. 1997, 1998, 2003). On the contrary, dopamine neurons will not typically respond to neutral stimuli, such as an arbitrary image or the ring of a bell. Yet, if the image is consistently followed by the delivery of juice then the dopamine response will slowly shift from responding to juice to responding to the mere presentation of the image (conditioned stimulus) (Schultz et al. 1997). This neuronal representation of a computation has been reliably demonstrated in both animals and humans.

How does learning (i.e., updating the value of the stimulus) happen, especially in more complex settings, where the outcomes of actions are stochastic? Taking bribery as an example (illustrated in Fig. 3), assume that a stimulus (being stopped by a police in a corrupt country) produces an action (option B, giving a bribe) that is usually followed by a reward (being allowed to continue the trip). Individuals who have had multiple direct or vicarious experiences of this incidence would develop a "habit" of giving bribes. To efficiently navigate such environments, the computational solution is to generate the difference between the realized and expected outcome, the so-called **Reward Prediction Error (RPE)** (Schultz et al. 1997, 2003; Sutton and Barto 1988). To adapt one's choices to a dynamic environment, decision makers could implicitly calculate this value and integrate it to update the values of alternative choices.

Neurobiological and behavioral evidence indicates that RPE indeed exists and is processed by the brain. Understanding how RPE signals are actually represented by corresponding neuronal responses helps us to further develop the computational model to decipher the learning process. For example, the responses of dopaminergic neurons resemble RPE signals and not just direct responses to rewards as it was initially suggested (see above). In fact, dopamine is more sensitive to *surprise* and unexpected outcomes than rewards, and this is the basis of learning. For instance, detailed single-unit recordings in monkeys (Bayer and Glimcher 2005; Schultz et al. 1997, 1998) show a striking resemblance to an RPE calculation, which is distinct from pure reward responses. Moreover, signals recorded employing functional magnetic resonance imaging (fMRI) in the human ventral striatum (an area rich in dopaminergic responses) on a trial-by-trial basis correlate with RPE signals to a great extent (Christopoulos and King-Casas 2015; Dreher and Burnod 2002; Haruno and Kawato 2006; McClure et al. 2003; Pessiglione et al. 2006; Schonberg et al. 2007). Furthermore, pharmacologically manipulating dopamine levels by administering either L-DOPA (augmenting dopaminergic responses) or Haloperidol (known to decrease dopaminergic responses) changed both the RPE-related signals in striatum as well as the speed of learning (Pessiglione et al. 2006).

In sum, model-free learning is based on RPE, and the computational models of this learning strategy could successfully predict *both* neuronal responses and behavioral choices. In conclusion, there is more than sufficient evidence that RPE signals are actually represented by corresponding neuronal responses.

Model-Based Learning and State Prediction Errors (SPE)

One concern with model-free learning is its flexibility and its potential failure to capture learning in more complex situations. Although model-free learning is simple and can be applied to many different problems (for instance learning that bribing is the way to avoid a ticket from police), this learning process is rather slow and cannot respond in a sophisticated manner to more demanding problems (for instance, learning to navigate a corrupted and complex system in another country). Thus, it may not capture the learning process in complex, hierarchical, or very volatile situations. Moreover, behavior can only change following direct or indirect experience of outcomes, and not by mental simulation (i.e., constructing in one's mind the possible outcomes of different actions in different situations, for instance, following a tree-type search method).

In contrast to model-free learning, in model-based learning, the decision maker aims at learning a mental model of the environment, which maps out the potential different *states* of the world and the relations of these different states—for instance, by employing a decision tree. A canonical (oversimplified) form is a sequential decision-making process: imagine a decision maker deciding on which button (I or II) to press depending on which room (A or B) she has entered; if she is in room A, pressing button I delivers a (desirable) can of orange juice, whereas pressing button II delivers nothing (because the button is broken); on the contrary, if she is in room B, button I is broken instead, whereas button II delivers a desirable can of orange juice. Take the bribery example (Fig. 3b)—a salesperson has worked in countries A and B; in country A, giving bribe leads to severe negative consequences, whereas in country B, one way to close deals is to bribe. Regardless how the salesperson acquires the state of the environment (by observation or direct experience), having the mental model would help him or her to achieve the desired outcome (close the deal) efficiently. As such, model-based learning is advantageous, as it can flexibly and quickly adapt the values[1] and probabilities depending on the different states.

[1] Notice that in this context, the term "value" refers to the magnitude of outcome and should not be confused with other interpretations such as social values etc.

Model-based learning might require ***mental simulation,*** whereas options can be represented in a forward (or tree-search) manner. This is in contrast with model-free learning, in which actions' values are learned in a direct manner, without the need to build a precise and explicit map (mental model) of the environment (Glascher et al. 2010). By performing mental simulation, individuals could also perform virtual 'mental searches' throughout the problem space to identify the solution. For example, in order to get permission to build a factory in Country C, a manager may simulate the following in her mind—"If I do not report the potential negative environmental impacts of the factory in the application, Office X may reject my application; yet, we can bribe minister Y who oversees Office X to approve the application." Thus, model-based learning is not only sensitive to past rewards but also allows the planning of future actions.

Interestingly, model-based learning is also supported by prediction error computations, namely the state prediction errors (SPE). Again, the decision maker compares the realized and predicted outcomes. Yet, the calculation of differences in the outcomes is derived from and used to update the values of ***internal cognitive map (or mental model)*** (see Fig. 3) and not the actions in a trial-and-error fashion. Model-based learning is also biologically supported. In brief, fMRI experiments combined with computational modeling suggest that the human brain supports both model-free and model-based learning, potentially using different biological pathways. While the model-free reward predictive error signals are typically mirrored in striatal areas, the model-based state predictive error signals are represented in lateral prefrontal areas and intraparietal sulcus (Beierholm et al. 2011; Daw et al. 2011; Deserno et al. 2015; Glascher et al. 2010).

Finally, learning processes can be supported by social information (for instance, observational learning, social comparisons and social influence). There is a host of literature using neuro-computational methods to address such social learning processes, which is out of the scope of the current paper (for some examples see (Aimone et al. 2014; Behrens et al. 2008; Burke et al. 2010; Christopoulos and King-Casas 2015; Chung et al. 2015; Lauharatanahirun et al. 2012; Suzuki et al. 2012).

Model-Based and Model-Free Learning Differences and Arbitration

The differentiation between model-free and model-based learning has been well demonstrated in experiments. For example, in a situation, where a rat has been trained to press a lever to obtain food, and it is well-fed to satiation: the model-based system will predict a small value to the lever press as the rat is not hungry anymore, but the habitual system will predict a (learned) large value to the lever press because it is a well-learned habit.

The key difference between the two learning strategies is that in model-free learning, individuals learn the values of actions in a direct manner, without building a precise and mental model of the environment (Glascher et al. 2010). This type of learning generates appropriate responses following stimulus-action-outcome associations. Model-free learning responses are usually established following previous experience and a lot of practice. Take the bribing situation as an example, after many repetitions, the action of paying a bribe (option B) becomes habitual. In contrast, model-based learning evaluates the actions and the consequential rewards of the actions based on current internal and external states (e.g., choosing to not pay a bribe because the situation occurs in a non-corrupt country), as illustrated in Fig. 3.

In fact, both model-free and model-based learning are needed in a complementary fashion. To elaborate, model-free learning is problematic, because it can be rigid and slow to change. In comparison, although model-based learning is more adaptive, it concurrently needs increased working memory—which might add a significant computational cost. Decision makers need to hold in memory a detailed cognitive map. Another major concern is that mental models in model-based learning can be inaccurate, which may result in suboptimal decisions and even problematic decisions in fast-changing environments. Therefore, decision making in adaptive environments need both model-free and model-based learning to complement each other.

As both learning strategies can be implemented in the brain, an important question is how are they combined to guide decisions? A similar question has also been raised in dual-system theories, as it is unclear how the decision maker chooses to adopt an automatic or a deliberative strategy. One suggestion (Daw et al. 2005; Doya 2002; Doya et al. 2002) is that choice is based on the certainty characterizing each strategy at a specific point of time: the strategy that seems to have higher reliability will be selected. This explanation seems to be generally supported by recent empirical findings (Glascher et al. 2010; Lee et al. 2014), suggesting the existence of a hybrid form of learning that combines input from both systems by forming a weighted average of the inputs from the two strategies (Deserno et al. 2015; Wunderlich et al. 2012). The negotiation between the two systems depends on the characteristics of the environment as well, especially during action selection. Although the exact processes through which options are chosen is not known, there is some evidence suggesting that the prefrontal areas seem to assign control to different systems. For instance, when the two systems have different "suggestions" to offer with respect

to the appropriate course of action, the prefrontal areas will choose which system to assign control to. In familiar environments, the habitual system will take over, as it can offer quicker responses with less cognitive demand. In situations that are unstable or new to decision makers, model-based systems might be preferred for decision makers to extract rules. Conflict between systems might be the reason for problematic behaviors, such as choice inconsistency and preference for short-term rewards [for the different systems see (Daw et al. 2005; Dayan et al. 2006; Loewenstein and O'Donoghue 2004)].

Learning Strategies and Moral Principles: Utilitarian and Deontological Reasoning

The two learning strategies could contribute to the understanding of the two dominant moral principles in moral philosophy, namely, utilitarianism/consequentialism, and deontology (Cushman et al. 2006; Greene 2009; Greene et al. 2009). We acknowledge that these two are not the only distinct moral principles, see for instance (Robertson et al. 2007). Utilitarianism/consequentialism focuses on the outcome of actions in moral decision making and moral judgment (for instance, kill somebody to save many others), whereas deontology focuses on the rightness or wrongness of the actions themselves (one should never kill somebody even when the action can save many others).

We propose that utilitarian choices are supported and driven by model-free learning, whereas deontological choices are supported by model-based learning. To elaborate, a utilitarian choice needs to calculate values and outcomes in a relatively mechanistic way, and choices are mostly context-independent and applied universally under this approach. On the contrary, deontological choices need to engage general rules and identify the violations of rules. Neurobiologically, given that prefrontal areas, to a great extent, support model-based learning, by extension they should also support deontological choices. In fact, research has found neurobiological evidence supporting that the prefrontal areas are engaged in moral decision making. For instance, experiments (Moll et al. 2002a) have shown stronger involvement of the ventro-medial prefrontal cortex (VMPFC) toward moral stimuli than non-moral stimuli, suggesting that the mere exposure to visual stimuli involving moral situations (such as war scenes) necessarily generates larger neural responses than emotionally evocative stimuli with less moral content (such as pictures of injuries). VMPFC is also sensitive to moral transgressions: VMPFC is selectively activated when moral stimuli or statements are contrasted with statements that are socially non-permissible but morally relatively neutral (such as licking a dirty toilet) (Moll et al. 2002b).

Importantly, VMPFC is more sensitive to moral scenarios that are personally involved than non-involved, resembling model-based (state dependent) learning and decision making. VMPFC activation is indeed higher in moral scenarios that are more personal (pushing an innocent person over the bridge) than impersonal (pushing a button to divert the train toward an innocent person) (Moll et al. 2001). Moreover, patients with VMPFC lesions were more likely to adopt utilitarian solutions than normal participants (Anderson et al. 1999). In that study, the main concern of the patients was the overall good and not the 'means' to achieve this overall good. These results suggest that VMPFC is critical for the acquisition (and possibly updating) of deontological (vs. utilitarian) moral reasoning and model-based (vs. model free) learning.

Moreover, there is some evidence that utilitarian choices could be subserved mostly by subcortical structures and thus possibly by model-free learning. That said, the causal direction based on lesions and patient data could be ambiguous (see the debate between (Greene 2007; Moll and de Oliveira-Souza 2007) and could run the risk of reverse inference. Therefore, the relationship between the deontological versus utilitarian moral system and the two learning strategies remains to be empirically addressed.

Valuation: A Multifactorial Assessment

A critical aspect of any dynamic learning process is *valuation*, the process through which individuals assess the outcomes received as well as make predictions for different actions. This is not a simple process, as it depends on various characteristics of the outcome, such as its temporal and probabilistic properties. Again, the properties of outcomes of actions have also been noted by theories on moral decision making. Jones' issue-contingent model (Jones 1991) suggested the following properties as important characteristics of moral decision making: magnitude (the sum of harms and benefits to both self and others), social consensus (and social factors overall), probability of effect, temporal immediacy (temporal proximity between action and outcomes), proximity (degree of similarity between decision maker and receivers of the action), and concentration of effect (a weighted average of the number of people affected and the harm induced to them).

There is an impressive correspondence between the concepts suggested by the issue-contingent model and the factors studied by DN. Specifically, the first four concepts have been very carefully examined and modeled by DN; the last two have not yet be examined. We believe that examining their neurobehavioral (i.e., behavioral responses along with their neural underpinnings) components would be a very exciting future research program, borrowing methods, and metrics well established in DN (Cohen and Aston-Jones 2005; Miyapuram and Pammi 2013; Sanfey 2007; Smith and Huettel 2010; Venkatraman 2013).

How does the brain integrate all these different values? Inspired by economics, DN suggests that the human brain acts 'as if' it attaches a value (called "utility") to each of the options (Glimcher et al. 2005).[2] Utility is contingent on different parameters, such as pure magnitude (i.e., how much money will be given), risk, temporal distance (outcome given now or later), effort, and even social factors, such as fairness (comparing one's outcome with the outcomes of others in the environment). In addition to these monetary and social factors, neuroscientists and philosophers suggest that actions could also carry moral utility which shape the valuation of choices (Tobler et al. 2008). Generally, utility transformations are a computational necessity for individuals to make decisions in a non-random fashion. Below, we summarize how each factor shaping valuation is encoded by the human brain.

Magnitude

Basic neuroscience results have convincingly identified brain areas that are sensitive to primary rewards (such as food or water) (de Araujo et al. 2003; Jiang et al. 2014; Kringelbach et al. 2003; Lawrence et al. 2012; Small et al. 2003). These areas include medial orbitofrontal cortex (OFC) and striatal areas, as well as anterior cingulate and anterior insula.

The same areas that can represent primary reinforcers are also sensitive to higher-order rewards (or punishments), such as money (Bayer et al. 2013; Izuma et al. 2008; Miyapuram et al. 2012; Osuch et al. 2009; Zink et al. 2004), or even social comparisons, social hierarchy, and reputation (Aharon et al. 2001; Fisher et al. 2005; Fliessbach et al. 2007; Harvey et al. 2010; Izuma et al. 2008; Lacey et al. 2011; Montague et al. 2014; Zink et al. 2008). Additionally, valuation responses are context-dependent, a phenomenon that has been behaviorally demonstrated (Christopoulos et al. 2011). For instance, they can be modulated by external information such as the brand of a soda (Coca-Cola or Pepsi) (McClure et al. 2004) or whether the very same odor was presented as a sweaty sock or smelly cheese (de Araujo et al. 2005).

Thus, the human brain's reward system can employ similar mechanisms and networks to signal both basic rewards (food, etc.) as well as more abstract concepts. Notice also that according to prospect theory (a popular decision-making theory, (Kahneman and Tversky 1979), losses are treated differently than gains (Fig. 4a). The main message of this graph is that "losses loom larger than gains," meaning that people prefer avoiding losses more than making gains. An additional important conclusion is that people are risk-averse when outcomes are (framed as) gains, whereas they are risk-seeking when outcomes are (framed as) losses.

Risk

Risk is a central concept in DN. Here we need to get formal and clarify some misconceptions about risk. First, contrary to popular belief, risk can be dissociable from probabilities—it can increase even by keeping the probabilities the same (Rothschild and Stiglitz 1970). For instance, a gamble offering \$400 or \$600 with equal probability (50 %) is less risky as compared to a gamble offering \$100 or \$900, again with equal probability. Another misconception is that the term risk is associated with the probability of a negative/lower outcome, or what is formally called *downside risk*. For instance, the statement, "now we increase the risk," might be translated as, "now we take a more perilous action," which is, formally, not the full story, as increasing risk makes sense by also increasing potential rewards (upside risk).

Also, risk is not a unitary concept. Besides dispersion, other important facets of risk are skewness and kurtosis. Skewness is of particular interest because it seems that risk-seeking persons (such as gamblers) can be attracted by positively skewed outcomes, whereas risk-averse persons can find those outcomes unattractive if not disturbing. For instance, bets that offer a small probability of a high outcome are particularly attractive to gamblers (Garrett and Sobel 1999).

From a neurological perspective, risk is encoded by a network of areas including the striatum, anterior insula, anterior cingulate, dorsomedial and lateral prefrontal cortex, and inferior parietal lobule (Huettel et al. 2005; Knutson et al. 2008; Kuhnen and Knutson 2005; Lauharatanahirun et al. 2012; McCoy and Platt 2005; Schultz et al. 2008). Skewness seems to have separate—to some extent—neural encoding—for details see (Burke and Tobler 2011; Symmonds et al. 2011). Risk attitudes and probability distortions could emerge even in conditions when decision makers do not need to make choices (Tobler et al. 2008).

Temporal Effect on Valuation of Reward and Cost

Common in business life, decision makers have to choose between short- and long-term interests. The value of an option can be modulated by the time the associated outcome is delivered sooner or later. Typically, these situations involve making a choice between small(er) rewards delivered sooner and large(r) rewards delivered later. In a bribery scenario, a person might choose to pay a public servant so as to receive a license more quickly. Termed as

[2] The concept of utility here is more generic than the way is used by "Utilitarian" approaches.

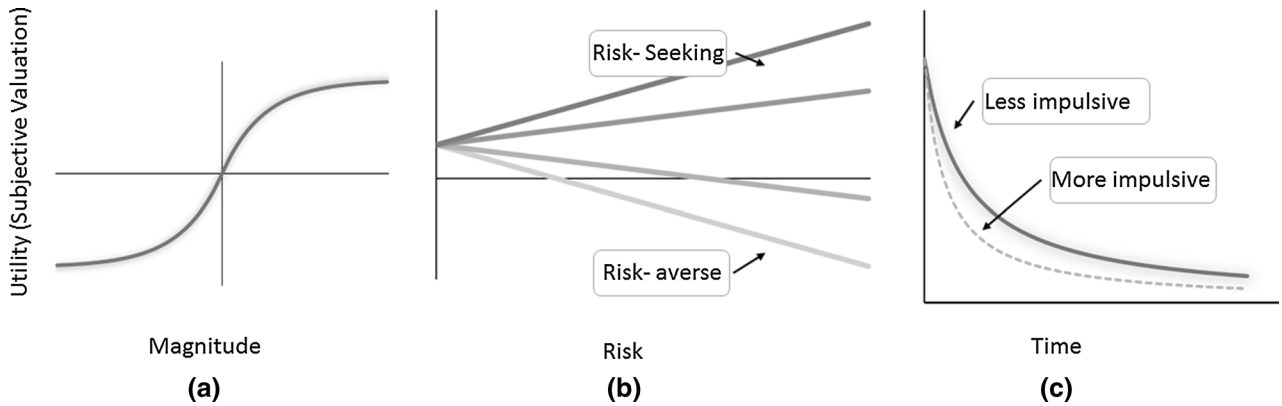

Fig. 4 Magnitude, risk, and time: their impact on valuation

temporal discounting, rewards are discounted as a function of the time of delivery, with rewards delivered later losing their value. Thus, if an outcome is delivered later, it would need to be of higher monetary value than those delivered sooner.

Temporal discounting is a well-studied phenomenon across life span (Green et al. 1994) and cultures (Du et al. 2002). Typically, children and adolescents have higher temporal discounting tendency as they prefer sooner rewards than adults (Green et al. 1994; Steinberg et al. 2009). Temporal discounting has proven to be a reliable characteristic of various clinical categories. For instance, cocaine users, smokers, individuals with ADHD, and pathological gamblers have higher discounting rates (Alessi and Petry 2003; Baker et al. 2003; Bickel et al. 1999; Heil et al. 2006; Paloyelis et al. 2010; Spinella et al. 2004). Overall, temporal discounting is associated with higher impulsiveness (see Fig. 4c).

It is still unresolved whether temporal discounting is a state or a trait. Studies on rats suggest that discounting is a variable trait, as the time preferences remain stable even after self-administered cocaine (Odum 2011). However, other studies imply that temporal discounting can be modulated by the delivery of amphetamines, both in humans and animals, suggesting that it can be a state (de Wit et al. 2002; Winstanley et al. 2003).

From a neuroscience perspective, impulsiveness can be driven by the competitive interaction of two different brain systems. Impulsive choices appear to be guided by subcortical limbic (and evolutionarily more primitive) structures, such as the amygdala, nucleus accumbens, and ventral pallidum. By contrast, choices favoring larger and delayed rewards seem to be subserved by fronto-parietal regions (called the Executive System). This distinction has been recently supported by neuroimaging evidence (McClure et al. 2004) and is also in agreement with the so-called "Somatic Marker Hypothesis" (Bechara and Damasio 2002; Bechara et al. 2005; Damasio 1996) (see definition in Table 1). Dorsolateral prefrontal cortex (DLPFC) areas seem to be instrumental in directing behavior toward future rewards, as it is an area associated with cognitive control. As Ballard and Knutson (2009) noted, representing a future reward necessitates inhibiting automatic responses, a function that is predominantly supported by DLPFC.

As morally risky behaviors can result in negative consequences, another important question is how time influences the valuation of ***cost***. This temporal issue was also discussed in the issue-contingent model such that the negative consequences of actions influence moral decision making (Jones 1991). For instance, releasing a drug that will cause long-term side effects is preferred over a drug that causes acute negative responses. An early investigation (Mischel et al. 1969) reported that college students preferred electrical shocks of equal intensity to be delivered earlier rather than later. Yates and Watts (1975) also found that when outcomes are negative, students were less impulsive. Later studies offered mixed results, with the main problem being that it is rather difficult to establish true negative outcomes in an experimental setting (Loewenstein 1988). For instance, deciding to postpone, a loss can be framed as an immediate gain. Past research has found that "gains are discounted more than losses," a phenomenon called the "sign effect" (Frederick et al. 2002; Loewenstein 1987, 1988; Thaler 1981a, b). Interestingly, the discounting factors of costs and rewards were correlated, suggesting a common mechanism.

Limited but solid support for the sign effect comes from neurological investigations. Bickel et al. (2009) found no differences when comparing gains and losses. Yet another study (Tanaka et al. 2014) examined neurobehavioral responses during inter-temporal decision making involving gains and losses. Participants who were more sensitive to the sign effect (discounting gains more than losses) had stronger insula responses to losses compared with insula responses to gains than did their less sensitive counterparts. Striatal responses were also higher for loss-related delays

compared with gain-related delays. Interestingly, this asymmetry seems to be advantageous and desirable: smoking, debt, and obesity have all been associated with the *lack* of this gain-loss asymmetry (Ikeda et al. 2010; Odum et al. 2002).

Overall, temporal discounting is a phenomenon founded across species, domains, frames (for gains or losses), and cultures. Yet, the intensity of this phenomenon varies a lot as a function of personality, cultural background, and domains. The effect of time on valuation is more complicated than what the issue-contingent model suggests (Jones 1991) and needs to be further investigated.

Implications and Directions for Future Research

In business environments, ethical considerations might directly contradict key performance indicators. Yet, how individuals integrate valuation factors (such as risk and reward) in their moral decision making remains to be further examined. Importantly, understanding the learning and updating mechanisms in moral decision making can help us identify inhibitory and facilitating factors, thus enabling us to design systems and processes (or 'decision architectures') to support or encourage desirable moral behaviors.

Rewards, Punishments, and Tackling Unethical Habits

The first obvious question is how reward and punishment impact and, critically, change moral decision making? Could rewards and punishments speed up adaptive responses? If so, through which learning strategy (model-free or model-based)? Is the saying, "Everyone has their price" true? If so, under what circumstances? Would financially incentivizing moral behavior (perhaps by offering a reward to whistle blowers or, generally, people who behave morally) have a positive or negative effect on behavior?

For punishment, although regulatory systems that detect and punish unethical behaviors are usually implemented in practice, there are questions to be answered regarding how punishment affects moral decision making. What kind of punishment (monetary, social, and legal) would be the most effective? What levels of punishment are appropriate? Is punishment alone adequate to inhibit immoral choices, or is there a need for rewarding actions as well?

As discussed earlier, research has shown that habitual responses are difficult to change. Could inducing model-based learning help in removing unwanted habitual responses? Recent neurobiological evidence supports this postulation (Gillan et al. 2015). A recent study found that individuals who employed model-based learning were less susceptible to the development of habits. It is also suggested by previous research that impairments in model-based learning are consistently associated with stress and compulsive disorders (Gillan et al. 2011; Voon et al. 2015). This could have direct implications in the way learning and training is structured: Identifying general rules of behavior and supporting development of concrete mental models pertaining to morality (as opposed to just reinforcing/punishing behaviors) could allow for more malleable thinking. Moreover, the training should best be implemented at places that are similar to real work spaces and conditions because learning could happen by associating states with external stimuli, even physical environments.

A critical question arises here. Would rewards or penalties actually deter people from violating basic rules of business ethics? The intuitive answer is affirmative, but research in behavioral economics has provided some counterintuitive examples in which punishment actually reinforces an undesirable behavior. In a classic study, Gneezy and Rustichini (2000) found that charging parents a small fine when they were late to pick-up their children from a child-care center actually increased the percentage of late pick-ups. One of the reasons was that the monetization of the behavior gave a different meaning to the violation (late pick-up). Any moral costs (for instance, the moral value of respecting the teacher's after work hours) were quickly dissolved when the transaction took on a monetary character. That is, the parents' monetary compensation for the teacher's extra hours made the behavior more morally defensible. In another example, the addition of incentives encouraging contributions in a public goods game resulted in less harsh punishment of free riding (Fuster and Meier 2010). As such, it is possible that punishments and penalties, especially in a monetary form, might have the opposite impact in a business context. Overall, incentives might significantly alter the way social interactions are framed, as well as the social norms that are supposed to guide moral behavior (Gneezy et al. 2011). Adding a monetary or other reward immediately removes the intrinsic and reputation-related (signaling) motivations that could support ethical actions. Take volunteering behavior as an example, people often participate for the intrinsic value of volunteering, including reputational effects. In an experiment (Ariely et la. 2009), incentives increased donations to charitable organizations only when they were given in private. When incentives were made public, the effects were reversed, suggesting that incentives given in public may undermine the donors' reputation, and thus reduce their motivation to donate.

Risk Perception

Risk complicates the decision making process and explains the necessity of a dynamic approach, as consequences of

options in real business settings are hard to predict, and thus the contingencies need to be learned and updated. For instance, it is uncertain whether auditing mechanisms will detect fraud or bribery. It is rather unknown how moral decisions change when uncertainty is introduced. This could be crucial for understanding regulatory decisions, such as when business management has to select between increasing rewards, increasing punishment, or increasing the efficiency of auditing mechanisms (which improves the probability of unethical actions being identified). Important questions could be what levels of perceived risk should be adopted? What is the correct balance between reward, punishment, and risk to optimize the control of unethical behavior in a cost-effective manner?

To take an example, risk and attitudes toward risk are critical factors in determining the decision to bribe or not. For instance, there is evidence (Abbink et al. 2002) that increasing the probability of being caught and punished for bribery decreases incidences of bribery. Organizations and governments employ regular auditing to prohibit bribery by increasing the chances that such actions would be discovered. In other words, the decision to bribe or not could be the outcome of risk assessment (and much less of a moral consideration). It is also less known how uncertainty impacts social decisions and morality. In many of the studies reported above, outcomes come with certainty. There is strong evidence from behavioral game theory that, while people might be willing to sacrifice personal self-interest for the public good, when the environment becomes stochastic (probabilistic), cooperation decreases and free riding increases (Gong et al. 2009; Kunreuther et al. 2009). Therefore, a comprehensive theory of moral decision processes should incorporate an understanding of how the brain assesses risk.

Conclusion

Individuals' moral decision making could be influenced by fast-changing environments, competing pressures from different stakeholders, and different practices observed or experienced in business locally and abroad. This dynamic process calls for more attention to understand the changing feature of individuals' moral decision making. In the present paper, we introduce two basic, mostly implicit, learning strategies (model-free and model-based learning), and develop a comprehensive framework to examine how individuals' learning strategies in moral decision making. Our framework builds on and complements previous theorizations that have already hinted at and called for the incorporation of more dynamic characteristics. As such, the proposed model is not antithetical to existing research. By integrating concepts, computational models and methodologies from DN, we delineate the key parameters in moral decision making and the corresponding neurological mechanisms.

One important mission of business ethics is to design environments, processes, and policies that ensure ethical behavior that is by and large self-motivated. What is often neglected is that this is a continuous learning process. More attention is thus necessary in understanding how people learn to make, adapt, and change moral decisions, and how decision factors could inhibit or boost updating processes.

We envision this paper as a facilitator of the communication between business ethics experts and neuroscientists. Research in moral decision making could further extend its understanding and the level of analysis by incorporating computational models featured in Decision Neuroscience. We believe that borrowing the models presented here and integrating them with existing approaches in business ethics could pave the way for innovative approaches to understanding moral decision making in organizational contexts and beyond.

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